

Sewage 2040: A Speculative Timeline

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Wastewater monitoring is a rapidly emerging technology that has largely gone under the radar. In 2025, 150 million Americans will have their wastewater surveilled. While the technology has rapidly developed since its wide-scale adoption during the COVID-19 pandemic, little to no attention has been paid to the potential ELSI (Ethical, Legal, and Social Implications) of its development. Below is a scenario that represents my best guess about what unregulated surveillance without intervention may look like.

Early 2020: The world sees its first glimpse of wastewater monitoring. However, most don't know of its existence.

Behind the numbers and statistics on the dashboards was a method most Americans had never heard of. While the public tracked COVID-19 through test results and hospitalization rates, a quieter surveillance system was taking shape in the sewers beneath their feet.

In February 2020, researchers in the Netherlands made a discovery that would reshape public health surveillance: they detected SARS-CoV-2 RNA in wastewater samples collected before the first clinical cases were confirmed in their communities. The virus, shed abundantly in feces by infected individuals, including those without symptoms, left traces in sewage that could be detected days or even weeks before traditional surveillance methods picked up an outbreak.

Similar findings emerged rapidly from Australia, Italy, Spain, and France. By April, U.S. news agencies were reporting on the research, though the coverage was buried beneath the avalanche of pandemic news. Scientists began collecting samples from wastewater treatment plants across the country, quantifying viral loads that represented the infections of thousands, sometimes millions, of people in a single test.

The advantages were immediately apparent. Wastewater surveillance could provide one to two weeks of advance warning before clinical case counts rose. It captured infections in asymptomatic individuals who would never seek testing. And it did so at a fraction of the cost of individual diagnostics: a single wastewater sample could represent an entire city's infection status.

What most people didn't realize was that this emergency measure was laying the foundation for something far more permanent and far-reaching than a pandemic response.

September 2020: The CDC launches the National Wastewater Surveillance System.

In September 2020, the Centers for Disease Control and Prevention formally established the National Wastewater Surveillance System (NWSS). The goal was to coordinate the fragmented local efforts that had sprung up across the country, creating a unified national infrastructure for tracking SARS-CoV-2 through sewage.

The system expanded with remarkable speed. Partnerships formed with state and local health departments. By December 2022, NWSS would cover approximately 47 percent of the U.S. population across 46 states, with over 1,000 sampling sites.

The system delivered on its promise. Wastewater data tracked COVID-19 waves with precision, and when the Omicron variant emerged in late 2021, some California sewersheds detected it before clinical cases were confirmed. Wastewater sequencing identified variant proportions that helped officials anticipate the characteristics of coming surges.

But in the rush to deploy, fundamental questions went unasked. Who would have access to the data? What other targets might be monitored? The surveillance infrastructure was built first. The guardrails would have to come later, if they came at all.

2021-2024: The pandemic recedes but surveillance does not.

As the acute phase of COVID-19 faded, wastewater surveillance did not retreat with it. Instead, the infrastructure built during the emergency began expanding to new targets.

The CDC added Mpox, Influenza A, and Respiratory Syncytial Virus (RSV) to the NWSS monitoring list. Researchers began testing for foodborne pathogens, antimicrobial resistance genes, and dozens of other biomarkers. The National Institute on Drug Abuse launched pilot programs in over 70 cities, measuring opioid metabolites, fentanyl, methamphetamine, and cocaine in sewage.

The stated purpose was compelling: real-time data on the overdose epidemic killing over 100,000 Americans annually. Wastewater monitoring could detect changes in drug supply, like the arrival of new synthetic opioids, before the effects of the drugs could take place.

As a result, the geographic precision began shifting. During the pandemic, most surveillance occurred at treatment plants serving hundreds of thousands of people. The data was anonymous. But as sampling moved "upstream" to lift stations, building outflows, and dormitory plumbing, the populations shrank. Population-level surveillance is starting to become individual surveillance.

2025: The present moment, a choice is still possible.

Today, approximately 150 million Americans live in communities where their wastewater is monitored. Most have no idea. There has been no national debate, no congressional hearing, no public referendum.

The legal landscape remains patchy. No federal law specifically addresses wastewater surveillance. HIPAA does not apply to population-level sewage data. The Genetic Information Nondiscrimination Act (GINA) of 2008, designed to prevent genetic discrimination in employment and insurance, was written for individually identifiable genetic test results, not aggregate sewage analysis.

The Fourth Amendment offers little protection. The Supreme Court's 1988 decision in *California v. Greenwood* held that individuals have no reasonable expectation of privacy in garbage left for collection. Lower courts have extended this reasoning to sewage, treating biological material flushed down toilets as "abandoned property" with no constitutional protection. No court has reconsidered this decision since the development of modern genetic analysis.

The CDC maintains voluntary ethical guidelines for NWSS, including a 3,000-person minimum reporting threshold and tools to remove human DNA from sequencing data. However, these are only recommended frameworks and not legal requirements. They don't bind state agencies, law enforcement, or private companies conducting their own surveillance.

We stand at a fork, what follows is one possible path: the path we take if nothing changes.

2026-2027: Regulation fails and the Wastewater Monitoring Industry explodes.

Representative Robert Garcia (D-CA) introduces the Wastewater Privacy Protection Act (WPPA) with bipartisan cosponsors: purpose limitations to prevent function creep, a 10,000-person minimum threshold, mandatory data destruction after 90 days, and a prohibition on law enforcement access without a warrant.

The opposition mobilizes. The wastewater analytics industry, now worth \$4.2 billion annually, funds a coalition including utilities, analytics companies, and law enforcement associations. The FBI warns the bill would "interfere with drug investigations." At the hearing, a CDC witness admits they have documented no actual harms, only "prospective" ones. The bill dies in committee, 28-24.

With federal regulation off the table, the commercial market matures. A startup called BioBot Analytics raises a \$180 million Series C, offering municipalities free monitoring in exchange for data rights they monetize through an analytics platform. By year's end, they cover 89 million Americans. Customers include insurance companies, real estate investors, and political consultants.

Marcus Williams, 34, Homeowner, Cleveland, OH

Marcus receives a letter: his homeowner's premium will increase 23 percent due to "updated neighborhood risk assessment". The representative mentions "environmental health indicators" but won't explain further. Marcus has lived here for eight years. He doesn't use drugs. He has no idea his street has been sampled weekly for two years, that data show elevated opioid metabolites three blocks away, or that this information was sold to insurers. He never consented. He was never informed. All he knows is that his costs are rising and his home value is declining, and no one will tell him why.

Civil rights organizations document "Redlining 2.0": majority-Black neighborhoods are three times more likely to be classified as "elevated risk" based on wastewater data than similar white neighborhoods. GINA's limitations have finally become apparent: the law covers individual genetic tests, not aggregate sewage analysis.

2028-2029: Continuous Fine Wastewater Monitoring Technology Develops.

MIT researchers announce SewerSense: low-cost sensors deployable throughout sewer networks for continuous, real-time monitoring. Previous surveillance required collecting samples and transporting them to labs. SewerSense sensors, costing under \$200 each, perform on-site molecular analysis using

CRISPR-based detection. A single city can deploy thousands, monitoring not just treatment plants but individual building outflows.

Police departments adopt it immediately. Laredo City Government installs 47 sensors, dividing the city into monitoring zones of 2,000-5,000 residents, flagging zones with elevated drug metabolites for increased patrols.

Maria Santos, 28, Riverside, CA

Maria has been stopped three times this month. Each time, a routine traffic stop, broken tail light, failure to signal. Each time, questions about drugs. She's a dental hygienist. She doesn't use drugs. She doesn't know her neighborhood has been flagged by wastewater analytics, that patrols have tripled, that officers have been instructed to find reasons to make contact with residents. She just knows something has changed. The police are everywhere. Her neighbors, mostly Black and brown, mostly working-class, are all talking about the same thing.

The ACLU sues, arguing that continuous fine monitoring violates the Fourth Amendment. The district court rules against them, citing *California v. Greenwood*: sewage is abandoned property. The court affirms 2-1, backing *Carpenter v. United States* on the grounds that wastewater data cannot identify individuals. The Supreme Court declines to hear the case. The precedent stands: wastewater surveillance needs no warrant.

2030-2031: Individual level Identification Wastewater Surveillance Technology is Developed.

A genomics startup called GenoSeq announces "Amplified Individual Recovery" (AIR): a technique for extracting and sequencing individual human DNA from wastewater samples. Combined with expanded genetic databases, consumer services, law enforcement collections, and newborn screening, it can identify individuals from community-level samples.

Congress tries again. Senator Don Bacon (D-NE) introduces the Biological Privacy Protection Act, amending GINA to cover environmental genetic data and requiring warrants for surveillance capable of individual identification. It passes the Senate 52-48. It dies in the House when the Committee chair refuses to schedule a hearing.

Then comes the FBI request that changes everything.

Dr. Sarah Park, 45, New York Public Health Laboratory Director

Dr. Park stares at the FBI email. They want archived wastewater samples from 2028, collected for COVID variant surveillance, to extract human DNA for a serial murder investigation using GenoSeq's AIR technique. The samples were collected under CDC protocols promising the data would never be used for law enforcement. But there's no law prohibiting it. The FBI has made clear: cooperate voluntarily, or they'll seek a court order. She thinks about the thousands of people whose genetic material sits in those freezers, people who flushed their toilets without knowing their DNA would be captured, stored, and eventually searched. She sends the samples.

The arrest makes national news: a serial killer caught through "innovative forensic techniques." What receives less attention: thousands of people's genetic material was searched without knowledge or consent, not because they were suspected of anything, but because they lived in the geographic area of interest. The defense attorney's motions are denied and the precedent is set.

2033-2040: Integration of Wastewater Monitoring into Everyday Society.

By 2033, wastewater surveillance has been fully integrated into infrastructure across the United States. Over 85 percent of Americans live in monitored sewersheds. Continuous sensor monitoring has replaced periodic sampling. Applications have multiplied: employers test workplace facilities for drug use, landlords test building outflows as lease conditions, schools track students, and insurance companies adjust premiums based on neighborhood biological profiles.

The Thompson Family, Suburban Atlanta, GA

David and Michelle are refinancing when their application is flagged for "property location risk factors." The loan officer mentions "community health indicators." David eventually learns that wastewater data showed elevated prescription metabolites, medications for depression and anxiety, incorporated into a "community wellness score" affecting property values and lending decisions. David doesn't take those medications. Neither does Michelle. Their teenage daughter was prescribed an antidepressant last year. They had no idea the chemical evidence was flowing through the sewer system, affecting their family's financial standing.

By 2040, the concept of biological privacy has become largely obsolete. GenoSeq's AIR technique has been incorporated into standard sensor platforms. Individual genetic identification occurs automatically, in real-time. Every person's health status, genetic predispositions, medication use, and behaviors are knowable through analysis of household waste, generated automatically, collected

without consent, retained indefinitely without any frameworks or guidelines protecting the average civilian.

Senator Cory Booker, 69, Chair, Senate Commerce Committee, 2038

Senator Booker reviews the draft of the Biological Privacy Restoration Act, the fifth major attempt at federal regulation since the failed WPPA of 2026. His staff has already flagged the problems. The industry coalition has pledged \$120 million to oppose it. Three members of his own committee have received six-figure contributions from wastewater analytics PACs. They argue that the bill's restrictions would eliminate 60,000 jobs in the first year alone.

The insurance industry has sent letters warning that restricting access to wastewater data would force premium increases of 15-20 percent across all policyholders. The law enforcement lobby argues the bill would "roll back a decade of progress against the opioid epidemic." Even the public health establishment is divided as the CDC quietly opposes provisions that would require destroying archived samples, arguing they represent an irreplaceable research resource.

Booker remembers voting against the WPPA in 2026. The harms were hypothetical then, and the industry was small enough to regulate. Now the harms are real but the industry has grown beyond Congress's reach. "We had a window," he tells his staff. "We missed it."

Senator Booker shelves the bill. There aren't enough votes. There never are anymore.

Twenty years after the first COVID-19 wastewater detection, what began as an emergency public health measure has become a comprehensive system for biological monitoring of the American population without barriers of privacy or consent. The window for meaningful reform has closed. The industry that grew in the regulatory vacuum has become too big to fail and too powerful to constrain.

